5.2.1 The nuclear model of the atom

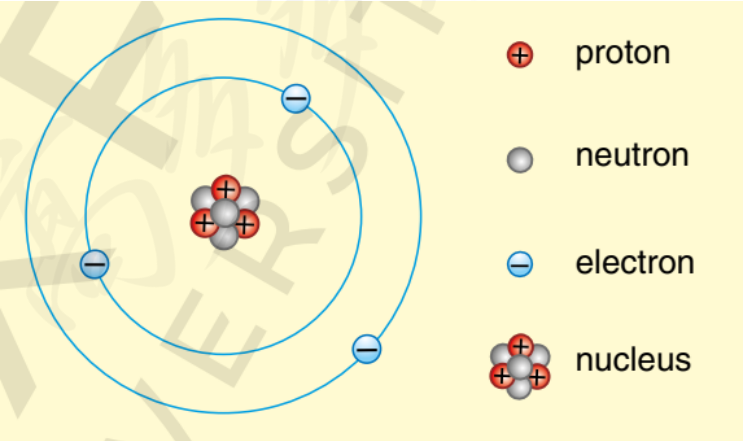
An atom has a very small positively charged nucleus surrounded by electrons. Most mass is concentrated in the nucleus and most atomic volume is empty space. Nuclei generally contain protons and neutrons. An ordinary hydrogen nucleus contains just one proton. A neutral atom has equal proton and electron counts and becomes an ion by losing or gaining electrons.

Atomic number and mass number

Atomic number is the number of protons and determines the element. Mass number counts protons plus neutrons, so neutron number is mass number minus atomic number. A neutral atom has as many electrons as its atomic number, but an ion does not. Mass number is an integer count, not the exact atomic mass in kg or u.

Isotopes

Isotopes are atoms of the same element with equal proton numbers but different neutron numbers. They therefore have equal atomic numbers and different mass numbers. Nuclear stability can differ: some are stable and others radioactive. Similar neutral-atom electron arrangements generally give similar chemical properties, but nuclear properties need not be identical.



Protons and neutrons form the nucleus; electrons occupy the surrounding region. The diagram is schematic and not to scale.

5.2.2 Activity and the becquerel

Activity A is the mean number of nuclear decays per second, measured in becquerels: 1 Bq is one decay per second. For N undecayed nuclei with decay constant λ, $A = \lambda N$. Divide decays per minute by 60 to convert to Bq. A detector usually records only part of the emitted radiation and includes background, so its count rate is not directly the total activity.

5.2.3 Activity and undecayed nuclei

Activity is the mean number of decays per second, equal to decay constant times the number of undecayed nuclei. For the same nuclide, more remaining nuclei mean greater activity. Activity is positive although the nucleus-count-time slope is negative. Detector count rate also depends on geometry and efficiency and must not be identified directly with activity.

$$A = kN$$

A: activity (Bq = s^{-1}); k: decay constant (s^{-1}); N: undecayed nuclei. Activity is positive; the decay rate of N is negative.

5.2.4 Half-life and external conditions

Half-life is the time for a large population of identical unstable nuclei, or its activity, to halve. It is characteristic of the nuclide and, in the course model, unaffected by temperature, pressure or chemical state. After n half-lives, the remaining fraction of nuclei and activity is $(\frac{1}{2})^n$.

5.2.5 Half-life and the remaining fraction

Half-life is the time for a large population of identical unstable nuclei, or its activity, to halve. Each half-life halves what remains; after two, one quarter remains. Divide elapsed time by half-life to find the number of half-lives, which may be non-integer.

Carbon-14 dating uses decay after an organism dies. While alive, it exchanges carbon with its environment and maintains an approximately steady C-14-to-stable-carbon ratio. After death, uptake stops while C-14 decays. Compare the sample's ratio or activity per unit mass with a living reference to estimate elapsed time.

The method suits once-living materials such as wood and bone. Very recent samples show too little reduction to measure accurately. Very old samples have low activity, making background radiation more significant.

$$N = N_0\left(\frac{1}{2}\right)^n$$

$$A = A_0\left(\frac{1}{2}\right)^n$$

$$n = \frac{t}{t_{1/2}}$$

n: elapsed number of half-lives, not necessarily an integer; N: undecayed nuclei; A: activity; subscript 0: initial value. No replenishment; single radionuclide.

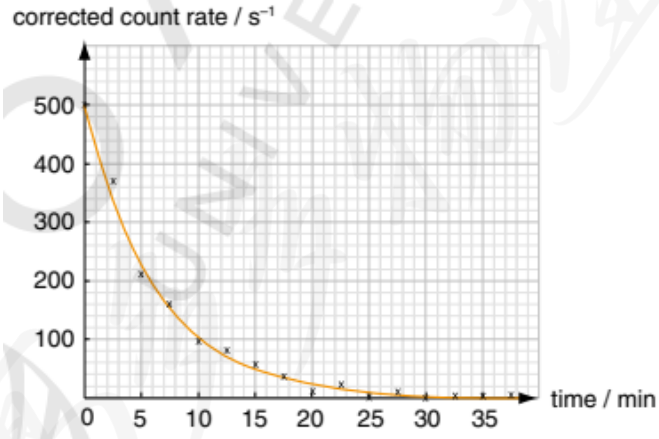
Exponential decay and the decay constant

For a single nuclide with no replenishment, undecayed nuclei and activity decrease exponentially because a fixed proportion of the current population decays per unit time. A larger decay constant means a shorter half-life. Match time units with reciprocal units of the constant. The half-life relation uses the natural logarithm, not base ten.

$$N = N_0 e^{-kt}$$
$$A = A_0 e^{-kt}$$
$$t_{1/2} = \frac{\ln 2}{k}$$

k: decay constant (s^{-1}); t: elapsed time (s); N: undecayed nuclei; A: activity (Bq); subscript 0: initial value. Single radionuclide with no replenishment.

$t_{1/2}$: half-life (s); k: decay constant (s^{-1}); ln: natural logarithm. Use matching time units.



Read the time for the count rate to halve and check whether the vertical-axis values have had background subtracted.